Day 082 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Represent and Graph Ratios

Standards: 6.RP.A.3a

Learning target: I can plot equivalent-ratio pairs and explain a point.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Put the first quantity on the horizontal axis and the second on the vertical axis. Build a table, label units, and plot ordered pairs. A point (x,y) describes both quantities together.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 082 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Represent and Graph Ratios

Standards: 6.RP.A.3a

Learning target: I can plot equivalent-ratio pairs and explain a point.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Put the first quantity on the horizontal axis and the second on the vertical axis. Build a table, label units, and plot ordered pairs. A point (x,y) describes both quantities together.

Worked example 1

For every 6 tickets sold, a club earns $2. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (18, 6).

1. Put the first quantity on the horizontal axis and the second on the vertical axis.

2. Build a table, label units, and plot ordered pairs.

3. Scale both quantities by each batch count

4. plot horizontal tickets and vertical dollars.

5. State the result in the form requested: (0,0), (6,2), (12,4), (18,6); 18 tickets earns $6.

Answer: (0,0), (6,2), (12,4), (18,6); 18 tickets earns $6

Worked example 2

A club earns $5 for every 5 tickets. Part A: plot ordered pairs for 0, 1, 2, and 6 batches (tickets, dollars). Part B: explain the meaning of (30, 30).

1. Put the first quantity on the horizontal axis and the second on the vertical axis.

2. For each batch count, multiply both 5 and 5 by that count.

3. Keep tickets horizontal and dollars vertical.

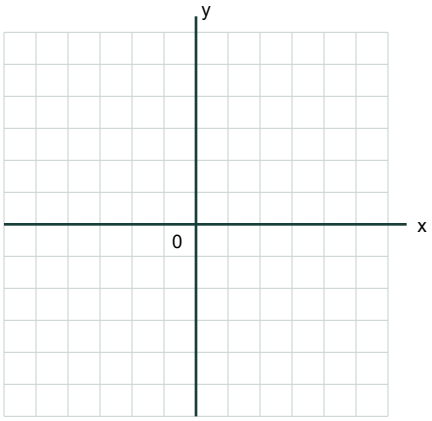
Answer: (0,0), (5,5), (10,10), (30,30); 30 tickets earn $30.

Day 082 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

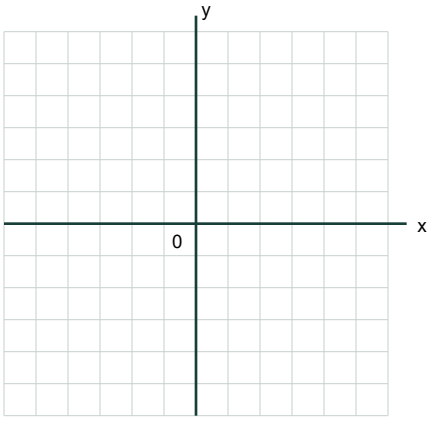
For every 6 tickets sold, a club earns $5. Make a table for 0, 1, 2, and 6 batches. Plot (tickets, dollars), label axes, and interpret (36, 30).



Label the scale and units.

02 / Constructed response

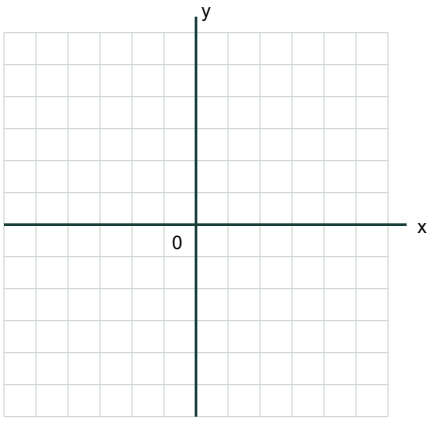
For every 2 tickets sold, a club earns $7. Make a table for 0, 1, 2, and 5 batches. Plot (tickets, dollars), label axes, and interpret (10, 35).



Label the scale and units.

03 / Constructed response

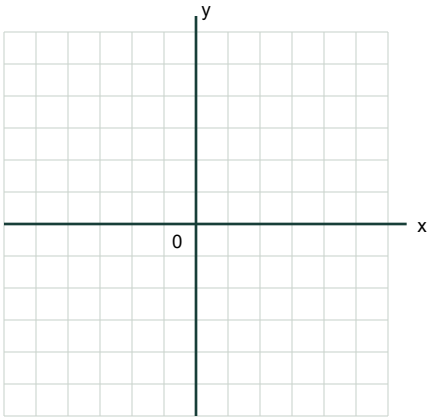
For every 7 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 2 batches. Plot (tickets, dollars), label axes, and interpret (14, 12).



Label the scale and units.

04 / Constructed response

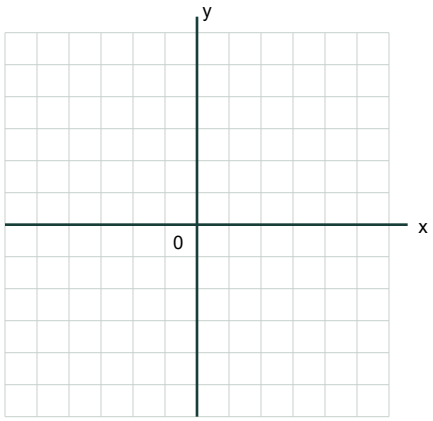
For every 2 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 4 batches. Plot (tickets, dollars), label axes, and interpret (8, 24).



Label the scale and units.

05 / Constructed response

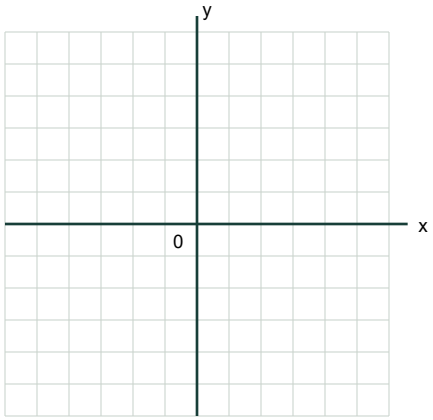
For every 2 tickets sold, a club earns $3. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (6, 9).



Label the scale and units.

06 / Constructed response

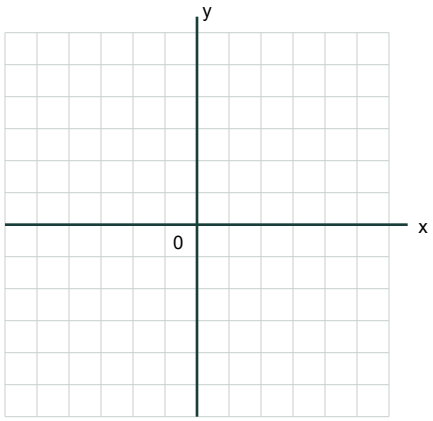
For every 5 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 2 batches. Plot (tickets, dollars), label axes, and interpret (10, 12).



Label the scale and units.

07 / Constructed response

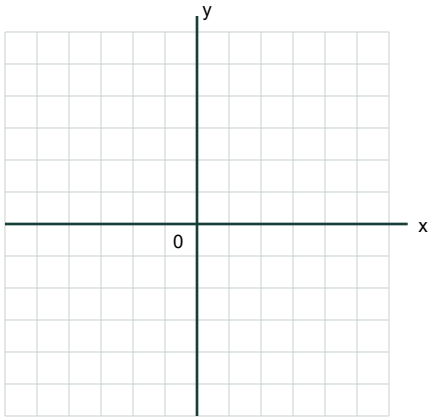
For every 2 tickets sold, a club earns $2. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (6, 6).



Label the scale and units.

08 / Table, graph, and interpretation

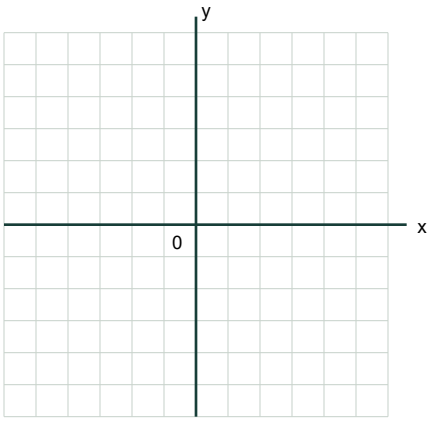
A club earns $2 for every 7 tickets. Part A: plot ordered pairs for 0, 1, 2, and 5 batches (tickets, dollars). Part B: explain the meaning of (35, 10).



Label the scale and units.

09 / Table, graph, and interpretation

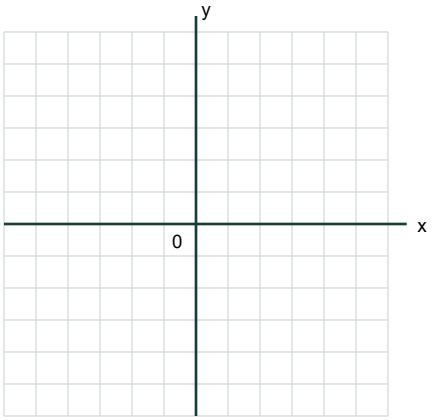
A club earns $6 for every 4 tickets. Part A: plot ordered pairs for 0, 1, 2, and 4 batches (tickets, dollars). Part B: explain the meaning of (16, 24).



Label the scale and units.

10 / Constructed response

For every 5 tickets sold, a club earns $4. Make a table for 0, 1, 2, and 6 batches. Plot (tickets, dollars), label axes, and interpret (30, 24).



Label the scale and units.

Day 082 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Represent and Graph Ratios

Standards: 6.RP.A.3a

Learning target: I can plot equivalent-ratio pairs and explain a point.

1. (0,0), (6,5), (12,10), (36,30); 36 tickets earns $30

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

2. (0,0), (2,7), (4,14), (10,35); 10 tickets earns $35

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

3. (0,0), (7,6), (14,12), (14,12); 14 tickets earns $12

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

4. (0,0), (2,6), (4,12), (8,24); 8 tickets earns $24

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

5. (0,0), (2,3), (4,6), (6,9); 6 tickets earns $9

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

6. (0,0), (5,6), (10,12), (10,12); 10 tickets earns $12

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

7. (0,0), (2,2), (4,4), (6,6); 6 tickets earns $6

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

8. (0,0), (7,2), (14,4), (35,10); 35 tickets earn $10.

For each batch count, multiply both 7 and 2 by that count. Keep tickets horizontal and dollars vertical.

6.RP.A.3a | Table, graph, and interpretation

9. (0,0), (4,6), (8,12), (16,24); 16 tickets earn $24.

For each batch count, multiply both 4 and 6 by that count. Keep tickets horizontal and dollars vertical.

6.RP.A.3a | Table, graph, and interpretation

10. (0,0), (5,4), (10,8), (30,24); 30 tickets earns $24

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response